Supplementary Materials for "RAPT: A Transformer-based Approach for Robust Symbolic Score-to-Performance Alignment"

This document contains supplementary materials for the paper titled "RAPT: A Transformer-based Approach for Robust Symbolic Score-to-Performance Alignment" submitted to ISMIR 2026.

1 Model Hyperparameters

For reproducibility, the full list of hyperparameters of the RAPT architecture and training pipeline are shown in Table 1, and the Optuna-tuned inference hyperparameters are shown in Table 2.

Symbol	**Description**	**Value**
F	Temporal resolution (quantization)	50 steps/s
W	Context window	37 (0.74 s)
d	Embedding dimension	256
L	Transformer layers	3
H	Attention heads	4
d_{ff}	Feed-forward dimension	512
K	Positional conv kernel	31
τ	InfoNCE temperature	0.07
p_{drop}	Note dropout probability	0.1
R	Sakoe-Chiba radius	50
θ_{keep}	Fusion IoU threshold	tuned per dataset
$\Delta_{\max}$	Max gap interpolation	1.5 s
Max epochs	Training budget upper bound	200 (15 for Batik)
Patience	Early-stopping patience	15 (999 for Batik)
Split seed	Train/val/test partition seed	1337
Weight seeds	Per-ensemble-member init seeds	{1001, 2002, 3003, 4004}

Table 1: Architectural and pipeline hyperparameters.

Configuration	α	β	θ_{keep}
Vienna 4x22	0.137	0.770	0.892
Batik plays Mozart	0.231	0.499	0.832
(n)ASAP	0.834	0.944	0.900
Joint training: ASAP+Vienna	0.076	0.695	0.852
Joint training: ASAP+Batik	0.332	0.804	0.868

Table 2: Optuna-tuned hyperparameters used for evaluation. The first three rows are in-domain configurations; the last two are joint training configurations used for cross-corpus evaluation.

2 Detailed Experimental Results

	ASAP	Batik	Vienna
Nakamura	0.9706 ± 0.0292	**0.9953 ± 0.0032**	0.9841 ± 0.0199
DualDTW	**0.9799 ± 0.0222**	0.9907 ± 0.0064	0.9887 ± 0.0064
GlueNote	0.9688 ± 0.0295	0.9800 ± 0.0140	0.9763 ± 0.0097
RAPT (Ours)	0.9780 ± 0.0226	0.9884 ± 0.0071	**0.9920 ± 0.0115**

Table 3: Comparison of RAPT against SOTA baselines on clean datasets (F1 Score: Mean ± Std).

Method	Vienna Δ	Batik Δ	ASAP Δ
Nakamura	-.225	-.003	-.001
DualDTW	**-.220**	-.001	-.001
GlueNote	-.221	-.002	-.002
RAPT (Ours)	-.223	**-.001**	**.000**
RAPT vs Baselines	$p > .05^{\dagger}$	$p > .05^{\dagger}$	$p < .001$

† Low test-set size (descriptive only).

Table 4: Degradation summary ($\Delta = \mathrm{F1}_{\mathrm{corrupt}} - \mathrm{F1}_{\mathrm{clean}}$). Negative values indicate performance loss under corruption. Wilcoxon signed-rank tests (Holm-Bonferroni corrected) compare RAPT against all three baselines.

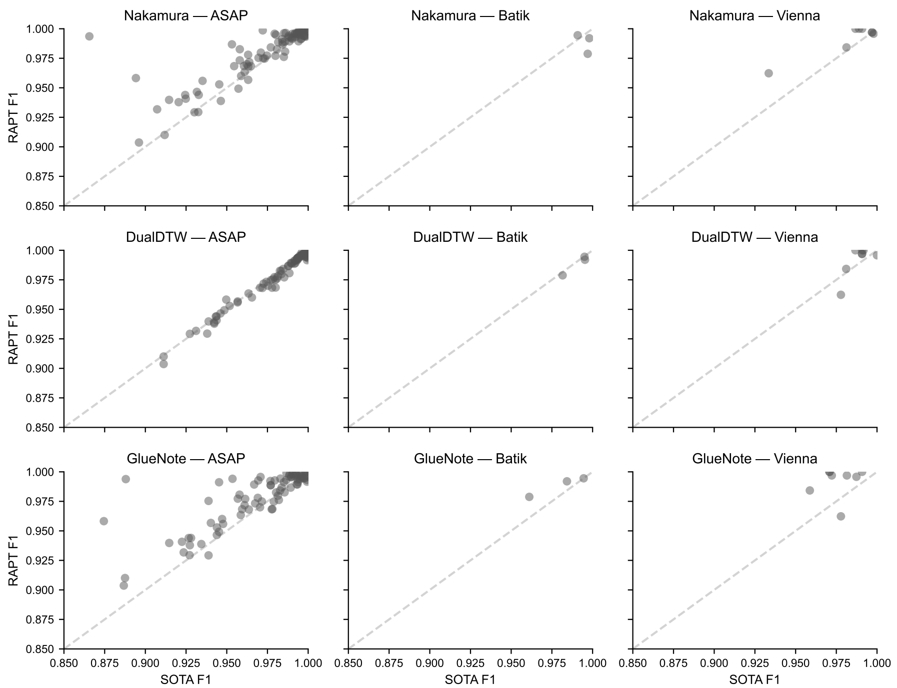

Figure 1: F1 scores of individual Pieces comparing SOTA and our model RAPT

3 Ablation Study

Ablation	Vienna	Batik	ASAP
RAPT (Canonical)	**.992**	**.988**	**.978**
Pure Transformer	.746	.891	.815
88-dim input	.991	.988	.977
No gap interpolation	.989	.989	.978
Mean pooling	.991	.989	.977
Max pooling	.992	.989	.977
IoU threshold=0	–	.978	–
Single seed	.991	.988	.975

Table 5: Ablation study on aggregate F1 scores (clean condition). The **Canonical RAPT** model serves as the baseline.

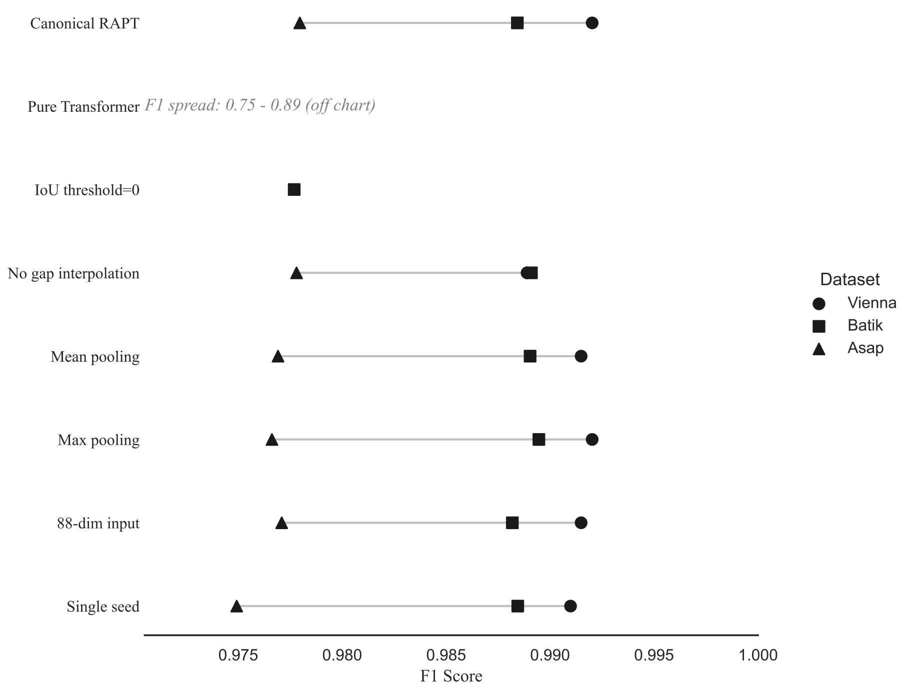

Figure 2: F1 scores of various configurations across datasets

4 Cross-Corpus Transfer

Method	**ASAP+Batik→Vienna** ($n = 88$)	**ASAP+Vienna→Batik** ($n = 36$)
Nakamura	0.976	0.981
GlueNote	0.974	0.973
DualDTW	**0.987**	**0.987**
RAPT	0.986	0.986

Table 6: Cross-corpus alignment F1: each ensemble is trained on two corpora and evaluated on the full held-out third corpus.

5 Compass Alignment Examples

Piece	RAPT	DualDTW	Nakamura	GlueNote
Chopin op. 38, p. 08 (V)	**1.000**	0.992	0.978	0.971
Chopin op. 38, p. 11 (V)	**1.000**	0.987	0.971	0.974
Schubert D.899/1 (A)	**0.957**	0.950	0.937	0.875
Haydn Sonata 31-1 (A)	**0.999**	0.997	0.999	0.989

Table 7: Per-piece F1 scores on a sample of pieces from Vienna 4x22 (V) and ASAP (A) highlighting where RAPT's contextual embeddings successfully disambiguate challenging regions.

6 Statistical Testing

Dataset	Comparison	Test	Statistic	p-value
ASAP	ANOVA (All)	F-test	3.4539	1.68e-02 *
	RAPT vs Nakamura	Wilcoxon/Mann-Whitney	671.5000	1.15e-06 ***
	RAPT vs DualDTW	Wilcoxon/Mann-Whitney	540.0000	4.59e-08 ***
	RAPT vs GlueNote	Wilcoxon/Mann-Whitney	545.5000	5.42e-08 ***
Batik	ANOVA (All)	F-test	1.1207	3.96e-01 ns
	RAPT vs Nakamura	T-test	-1.0817	3.92e-01 ns
	RAPT vs DualDTW	T-test	-2.5705	1.24e-01 ns
	RAPT vs GlueNote	T-test	1.6551	2.40e-01 ns
Vienna	ANOVA (All)	F-test	1.9218	1.49e-01 ns
	RAPT vs Nakamura	T-test	2.1908	6.46e-02 ns
	RAPT vs DualDTW	T-test	1.0215	3.41e-01 ns
	RAPT vs GlueNote	T-test	2.9393	2.17e-02 *

Table 8: Statistical significance testing across standard datasets. Includes One-way ANOVA followed by pairwise tests (Wilcoxon/Mann-Whitney for non-normal distributions, T-test otherwise). Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns (not significant).

7 Additional Discussion

7.1 Vienna 4x22: A Stress Test for Alignment

The dramatic 0.22 F1 drop on Vienna under corruption, shared by all four methods, deserves separate examination. Vienna 4x22 [?] contains only 4 unique compositions, each performed by 22 pianists. The limited score-level diversity means that test pieces are structurally very similar to training pieces in the clean setting; under corruption, the room for the alignment to use score-level disambiguation cues collapses, and every method drops to the ~0.77 F1 range.

This is a corpus-level property of Vienna, not a method-specific weakness: the closest baseline (DualDTW) shows comparable degradation. We hypothesize that performance on Vienna corrupted is bounded by the maximum amount of structural information that can be inferred from heavily-perturbed performances of compositionally homogeneous works.